Complete Robotics & High-Salary Engineering Learning Path

This document is a structured roadmap for Aryan Kumar Singh, a Mechanical Engineering student, to transition into a high-value Robotics and Autonomous Systems Engineer capable of targeting ■20–25+ LPA roles through strong technical skills, real-world projects, and deep-tech expertise.

1. The Main Goal

- Become a complete systems engineer, not only a mechanical engineer.
- Develop skills across robotics, embedded systems, AI, and automation.
- Focus heavily on practical implementation and project building.
- Target product companies, robotics startups, and deep-tech organizations.

---

2. Exact Skill Priority Order

- Linux
- Python
- C++
- Git & GitHub
- Electronics basics
- ESP32 / STM32
- ROS2
- Control systems & PID
- OpenCV
- Embedded Systems
- SLAM & Navigation
- AI & Computer Vision

---

3. First 3 Months Learning Path

- Learn Linux terminal usage and commands.
- Master Python basics deeply.
- Understand variables, loops, functions, OOP concepts.
- Start Git and GitHub.
- Learn C++ fundamentals.
- Build small ESP32 projects.
- Understand sensors and motors.

---

4. Beginner Projects

- LED automation system
- Bluetooth controlled car
- Obstacle avoidance robot
- Servo motor controller
- Line follower robot
- Basic IoT dashboard

---

5. Months 4–8 Learning Path

- Learn ROS2 architecture and communication.
- Understand nodes, topics, publishers, subscribers.
- Study PID control.
- Learn OpenCV for computer vision.
- Understand motor drivers and encoders.
- Study LiDAR and ultrasonic sensors.
- Learn basic path planning.

---

6. Intermediate Projects

- Autonomous rover
- Maze-solving robot
- Object detection robot
- Camera tracking robot
- Self-balancing robot
- Mapping robot

---

7. Months 8–15 Advanced Learning

- Learn SLAM deeply.
- Understand sensor fusion.
- Learn navigation stacks in ROS2.
- Study AI integration in robotics.
- Understand drone and autonomous systems.
- Contribute to open-source robotics projects.

---

8. Advanced Projects